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15 1. Dynamic Arrays (ArrayList) -> storing historical price series

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

haris@Hariss-MacBook-Pro ~ % /usr/bin/env /Users/haris/Library/Java/JavaVirtualMachines/openjdk-26.0.1/Contents/Home/bin/java --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp /private/var/folders/01/64c14s_d3t9b3mswtr937jhm0000gn/T/vscodesws_00a46/jdt_ws/jdt.ls-java-project/bin FinancialForecasting ^C
haris@Hariss-MacBook-Pro ~ % /usr/bin/env /Users/haris/Library/Java/JavaVirtualMachines/openjdk-26.0.1/Contents/Home/bin/java --enable-preview -XX:+ShowCodeDetailsInExceptionMessages -cp /private/var/folders/01/64c14s_d3t9b3mswtr937jhm0000gn/T/vscodesws_00a46/jdt_ws/jdt.ls-java-project/bin FinancialForecasting

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FINANCIAL FORECASTING - DSA DEMONSTRATION
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1) Price Ledger (Linked List):
Day 1: 102.50 -> Day 2: 104.00 -> Day 3: 103.20 -> Day 4: 105.80 -> Day 5: 107.10 -> Day 6: 106.40 -> Day 7: 108.90 -> Day 8: 110.20 -> Day 9: 109.50 -> Day 10: 111.80 -> Day 11: 113.00 -> Day 12: 112.40 -> Day 13: 114.60 -> Day 14: 116.10 -> Day 15: 115.70

2) 3-Day Simple Moving Average (Sliding Window):
[103.23, 104.33, 105.37, 106.43, 107.47, 108.5, 109.53, 110.5, 111.43, 112.4, 113.33, 114.37, 115.47]

3) 3-Day Weighted Moving Average:
[103.35, 104.63, 106.02, 106.53, 107.77, 109.13, 109.63, 110.77, 112.02, 112.5, 113.6, 114.98, 115.65]

4) Exponential Moving Average (alpha = 0.3):
[102.5, 102.95, 103.03, 103.86, 104.83, 105.3, 106.38, 107.53, 108.12, 109.22, 110.36, 110.97, 112.06, 113.27, 114.0]

5) Prefix Sum Query -> Average price, Day 5 to Day 10: 108.98

6) Daily % Returns sorted (Merge Sort):
[-0.77, -0.65, -0.64, -0.53, -0.34, 1.07, 1.19, 1.23, 1.31, 1.46, 1.96, 2.1, 2.35, 2.52]
Best single-day return : 2.52%
Worst single-day return: -0.77%

7) Binary Search -> Price on Day 10: 111.8

8) Greedy Algorithm -> Max profit (single buy/sell): 13.6

9) Linear Regression Forecast for next 5 days:
Day 16 -> Predicted Price: 117.34
Day 17 -> Predicted Price: 118.33
Day 18 -> Predicted Price: 119.32
Day 19 -> Predicted Price: 120.32
Day 20 -> Predicted Price: 121.31

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END OF DEMONSTRATION
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haris@Hariss-MacBook-Pro ~ %
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